

Arithmetic and complexity

Border rank of the 3×3 matrix product

Difficulty: extreme **Importance:** interesting to the community

Status: Open

Rating rationale: Extreme because exact border rank remains unknown even for this fixed matrix-product tensor; community importance reflects degenerating bilinear algorithms and the geometry behind fast multiplication.

Topic: approximate bilinear algorithms

Problem statement

In $\mathbb{C}^9 \otimes \mathbb{C}^9 \otimes \mathbb{C}^9$, let

$$M_3 = \sum_{i,j,k=1}^3 e_{ij} \otimes e_{jk} \otimes e_{ki}.$$

The tensor rank $R(T)$ is the minimum number of pure tensors in an exact sum for T . The border rank $\underline{R}(T)$ is the least r such that T is a Euclidean limit of tensors of rank at most r . Determine $\underline{R}(M_3)$. The use of limits makes this a different invariant from AC-02.

Why it matters

Degenerating bilinear algorithms underlie asymptotic improvements in matrix multiplication.

References and status

A. Conner, A. Harper, J. M. Landsberg, *New lower bounds for matrix multiplication and $\det_3$* , Theorem 1.1, proves $\underline{R}(M_3) \geq 17$. J. Alman and B. Li, *Asymptotic Rank Speedup Theorems, Revisited* (2026), §1, explicitly identifies this exact border rank as open. Searches for “3x3 border rank matrix multiplication 2026” found no exact determination. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked [Alman–Li, §1](#), and searched for a 2026 exact border-rank determination of the 3×3 multiplication tensor. The May 2026 source explicitly lists this value as open. Neither its asymptotic-rank improvements nor an exact-rank algorithm determines the border rank. No full resolution was located; the previously cited lower bound remains a bound, not an exact answer.